

Multimodal Emotion Recognition using HuBERT and BERT-based Multimodal Learning

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1. ABSTRACT

Emotion recognition is an important research area in affective computing and Human-Computer Interaction (HCI), enabling intelligent systems to understand and respond to human emotional states. Traditional emotion recognition systems primarily rely on handcrafted acoustic features and unimodal machine learning approaches, which often suffer from limited contextual understanding and poor robustness across varying emotional patterns.

This work proposes a transformer-based multimodal emotion recognition framework integrating self-supervised speech representation learning and contextual text modelling for robust emotion classification. The proposed framework evaluates three experimental settings: speech-only emotion recognition, text-only emotion recognition, and multimodal fusion-based emotion recognition. For speech representation learning, a pretrained HuBERT encoder is employed to extract rich acoustic and paralinguistic embeddings from raw speech signals. For textual representation learning, a BERT-based contextual encoder is utilized to generate semantic sentence embeddings from transcript text. The learned speech and text representations are subsequently integrated through feature-level fusion for multimodal classification.

The proposed system is evaluated using the Toronto Emotional Speech Set (TESS) dataset containing seven emotional categories. Experimental analysis demonstrates that speech-based transformer embeddings provide highly discriminative emotional representations with strong inter-class separability, while the text-only pipeline exhibits comparatively weaker performance due to limited semantic emotional richness within the transcripts. The multimodal fusion framework effectively combines complementary information from both modalities while preserving dominant emotional characteristics from speech representations.

In addition to classification performance evaluation, representation-level analysis is conducted using embedding visualization techniques such as t-SNE and PCA to study emotion cluster separability across temporal modelling, contextual modelling, and fusion representations. Extensive error analysis and emotion-wise evaluation are also performed to analyze model behavior, modality contribution, and fusion effectiveness.

The experimental findings demonstrate the effectiveness of transformer-based multimodal learning for emotion recognition while also highlighting the importance of modality informativeness in multimodal affective computing systems.

2. INTRODUCTION

Emotion recognition is an important research area in artificial intelligence and human-computer interaction. Emotions can be inferred from multiple modalities such as speech, facial expressions and text. In this project, a multimodal emotion recognition system was developed using speech-only, text-only and fusion-based approaches.

The objective of this work is to compare the effectiveness of:

1. Speech-only emotion recognition
2. Text-only emotion recognition
3. Multimodal fusion-based emotion recognition

The system was implemented using deep learning architectures with pretrained transformer-based feature extractors.

3. DATASET

3.1. Toronto Emotional Speech Set (TESS)

The Toronto Emotional Speech Set (TESS) dataset was used for this project.

Dataset Characteristics

- Contains emotional speech recordings
- Recorded by professional female speakers
- Studio-quality clean recordings
- Balanced emotion distribution
- Includes corresponding textual content and emotion labels

Emotion Classes

The dataset contains the following emotions:

Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, Sad

Transcript Generation

The filenames in TESS contain the spoken word information. Transcripts were generated automatically using filename parsing.

Example:

Filename	Generated Transcript
OAF_dog_angry.wav	Say the word dog
YAF_chair_happy.wav	Say the word chair

4. SYSTEM ARCHITECTURE

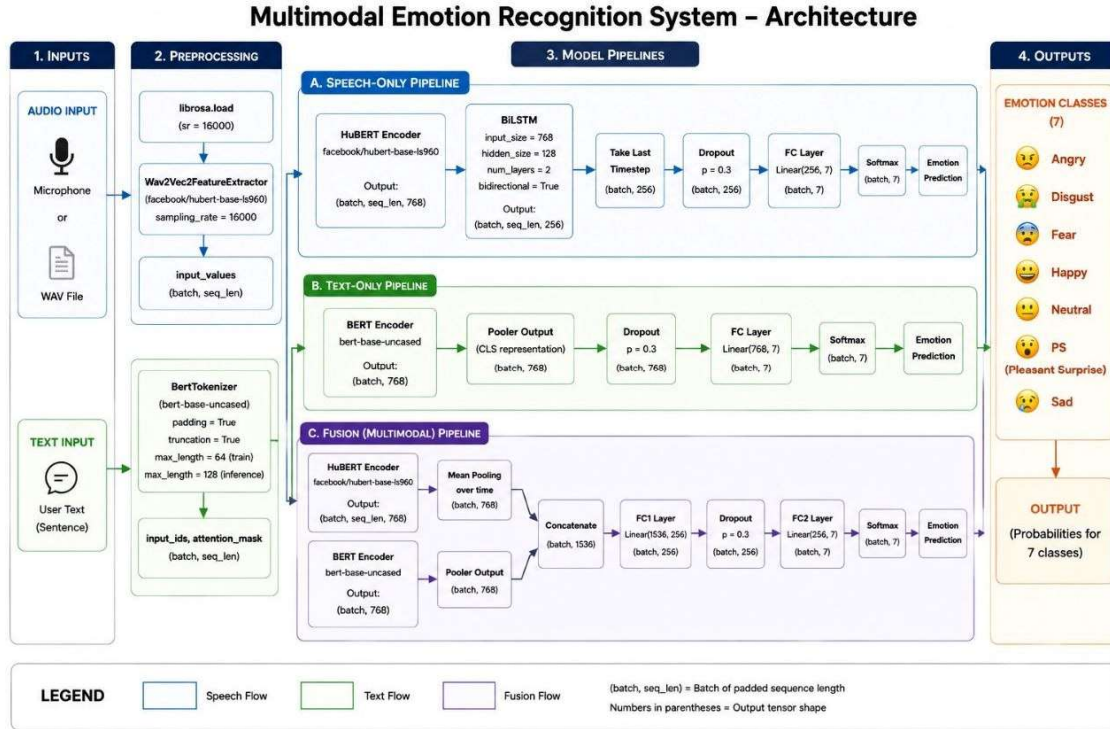


Figure 1: Overall multimodal emotion recognition system architecture.

The complete system consists of the following blocks:

1. Preprocessing
2. Feature Extraction
3. Temporal/Contextual Modelling
4. Fusion
5. Classification

5. SPEECH-ONLY PIPELINE

5.1. Architecture

Layer	Configuration
Input	768-dimensional HuBERT embedding
BiLSTM	128 hidden units
Dropout	0.3
Dense	128 neurons
Output	7 emotion classes

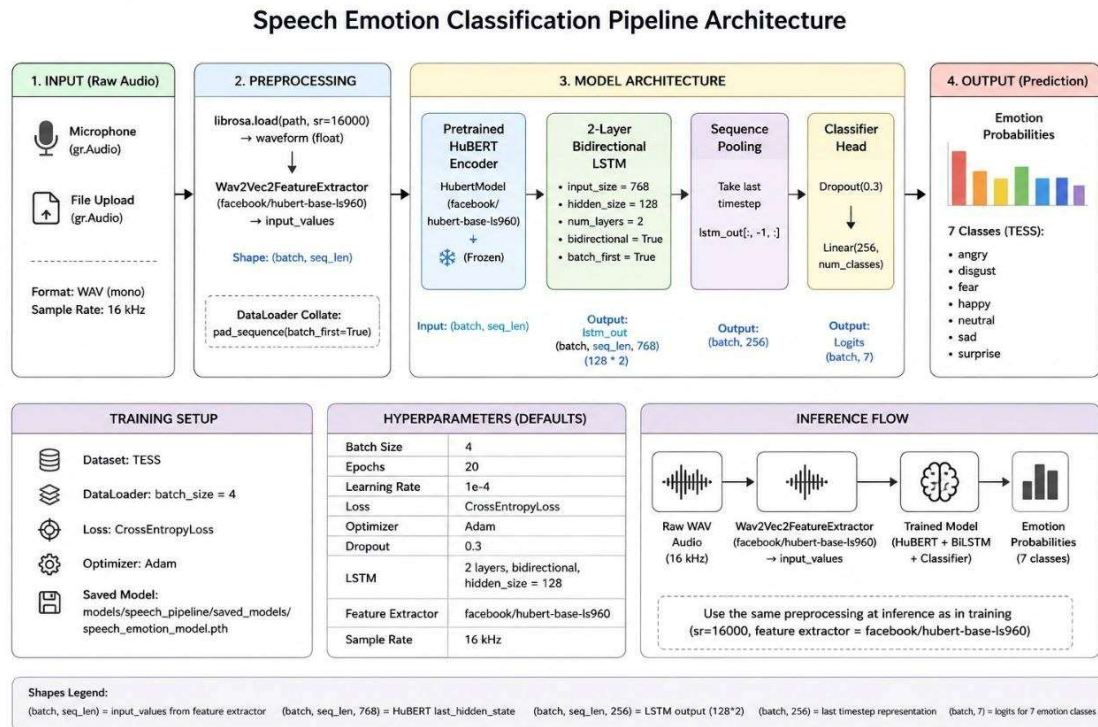


Figure 2: Speech-only emotion recognition pipeline architecture.

5.2. Preprocessing

The following preprocessing operations were performed on speech signals:

- Audio resampling to 16 kHz
- Stereo-to-mono conversion
- Fixed sampling rate normalization
- Tensor conversion for transformer processing

Why 16 kHz?

HuBERT was pretrained using 16 kHz speech signals. Using the same sampling rate ensures compatibility with pretrained weights.

5.3. Speech Feature Extraction

Architecture Used: HuBERT

HuBERT (Hidden Unit BERT) was used as the speech feature extractor.

Why HuBERT?

HuBERT was selected because:

- It captures rich acoustic and prosodic features
- It is pretrained on large speech corpora
- It performs extremely well on speech understanding tasks
- It extracts high-level emotional representations
- It generalizes better than handcrafted features such as MFCCs

Feature Extraction Process

1. Speech waveform is passed through HuBERT
2. Hidden representations are extracted
3. Mean pooling is applied over time dimension
4. A fixed 768-dimensional embedding is generated

Cached Embeddings

The extracted embeddings were stored as:

- `hubert_features.npy`

This significantly reduced computation time during training.

Tradeoffs and Limitations

Although HuBERT provides highly discriminative speech representations, it also introduces several computational tradeoffs.

Firstly, transformer-based speech encoders are computationally expensive compared to traditional handcrafted acoustic feature extraction techniques such as MFCCs. Feature extraction using HuBERT required significantly higher inference time and GPU memory consumption due to the large pretrained transformer architecture.

Secondly, the extraction process produced high-dimensional embeddings (768 dimensions), increasing storage requirements during cached embedding generation.

Additionally, the TESS dataset is relatively small and recorded under controlled studio conditions. While HuBERT generalized effectively on this dataset, performance on real-world noisy emotional speech may differ significantly.

Another important limitation is speaker dependency. Since TESS contains recordings from a limited number of speakers, random train-test splitting may lead to optimistic evaluation performance due to speaker-specific acoustic patterns being shared across training and testing sets.

5.4. Temporal Modelling

Architecture Used: BiLSTM

A Bidirectional LSTM (BiLSTM) was used after HuBERT embeddings.

Why BiLSTM?

BiLSTM was chosen because:

- Speech emotions evolve temporally
- BiLSTM captures sequential dependencies
- Bidirectional processing improves contextual understanding
- Lightweight compared to full transformer fine-tuning

5.5. Speech Classification

The final dense layer predicts one of the seven emotion classes using softmax activation.

Loss Function

Sparse categorical cross-entropy was used.

Optimizer

Adam optimizer was used.

5.6. Confusion Matrix analysis

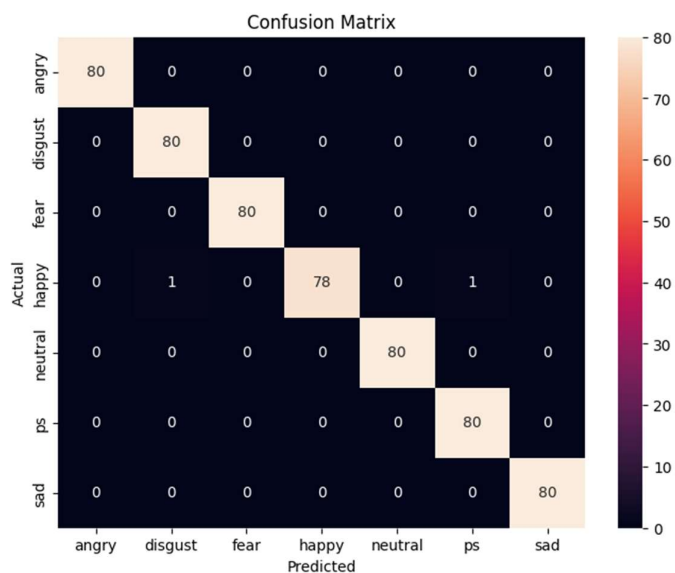


Figure 5: Speech-only pipeline confusion matrix.

Analysis:

The confusion matrix for the speech-only pipeline demonstrates strong diagonal dominance, indicating highly accurate emotion classification across all seven classes.

Most emotion categories were classified perfectly, highlighting the effectiveness of HuBERT embeddings in capturing acoustic emotional characteristics such as pitch, tone, intensity and prosodic variations.

Minor confusion was observed between pleasant surprise and acoustically similar emotions such as happy and fear. This overlap is expected because these emotions often share elevated pitch and energetic speech patterns.

Overall, the confusion matrix confirms that speech representations provide highly discriminative emotional information within the TESS dataset.

5.7. Results

Emotion	Precision	Recall	F1-Score	Support
angry	1	1	1	80
disgust	0.987654	1	0.993789	80
fear	1	1	1	80
happy	0.987654	1	0.993789	80
neutral	1	1	1	80
ps (pleasant_surprise)	1	0.975	0.987342	80
sad	1	1	1	80
Accuracy			0.996429	560
Macro Avg	0.996473	0.996429	0.996417	560
Weighted Avg	0.996473	0.996429	0.996417	560

Observations :

This model achieved the highest performance among all experimental settings.

The speech-only pipeline produced:

- Accuracy: 0.996429
- Weighted F1-score: 0.996417

The near-perfect performance indicates that emotional information is strongly encoded within the acoustic characteristics of speech. The HuBERT encoder successfully captured:

- pitch variations
- speaking intensity
- rhythm
- prosodic cues
- vocal energy patterns

The TESS dataset contains studio-quality emotional recordings with clearly expressed emotions, which further contributed to strong class separability.

6. TEXT-ONLY PIPELINE

6.1. Architecture

Layer	Configuration
Dense	256 neurons
Dropout	0.3
Dense	128 neurons
Output	7 classes

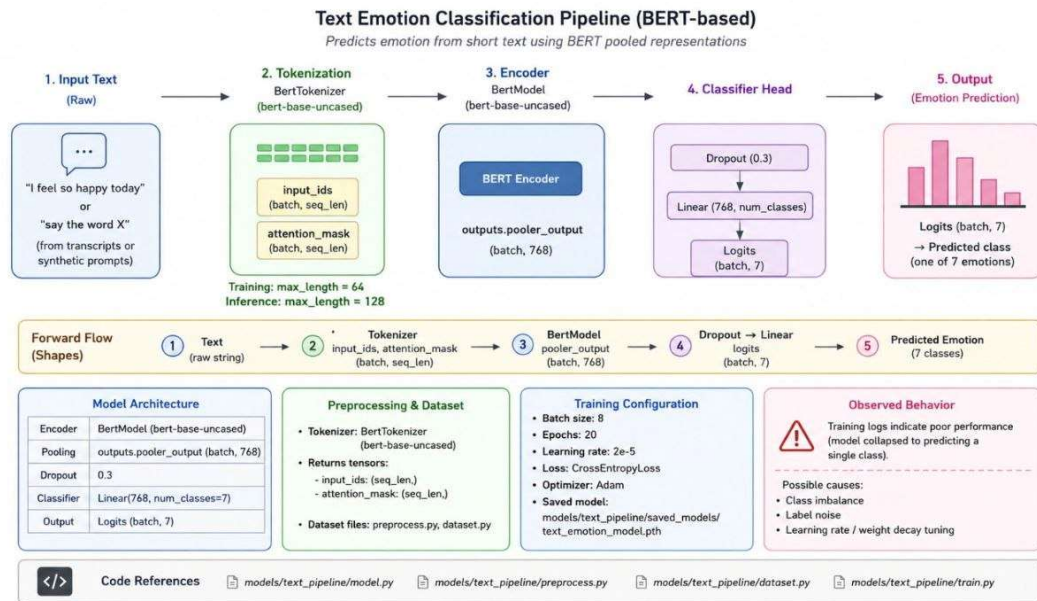


Figure 3: Text-only emotion recognition pipeline architecture.

6.2. Text Preprocessing

The following text preprocessing steps were performed:

- Filename-based transcript extraction
- Lowercasing
- Tokenization using BERT tokenizer
- Padding and truncation

Example

Original Filename	Transcript
OAF_dog_happy.wav	Say the word dog

6.3. Text Feature Extraction

Architecture Used: BERT

BERT-base-uncased was used for contextual text representation learning.

Why BERT?

BERT was selected because:

- It captures contextual token relationships
- It generates high-quality sentence embeddings
- It is pretrained on large text corpora
- It performs well for NLP tasks

Feature Extraction Process

1. Transcript tokenization
2. BERT contextual encoding
3. Pooler output extraction
4. 768-dimensional embedding generation

Cached Embeddings

Embeddings were stored as:

- bert_features.npy

Tradeoffs and Limitations

BERT provides powerful contextual semantic representations; however, its effectiveness strongly depends on the semantic richness of the input text.

In the TESS dataset, transcripts are synthetically generated from filenames and contain lexically neutral phrases such as “Say the word dog” or “Say the word chair.” As a result, BERT embeddings across different emotion classes become highly similar, limiting emotion separability.

Furthermore, transformer-based language models require substantial memory and computational resources during embedding extraction despite using frozen pretrained weights.

Another limitation is that the text modality in this dataset lacks emotional semantics, making it unsuitable as a standalone modality for robust emotion classification.

6.4. Contextual Modelling

Architecture Used: Dense Neural Network

Since BERT already performs contextual modelling internally through self-attention, a lightweight dense classifier was sufficient.

6.5. Confusion Matrix Analysis

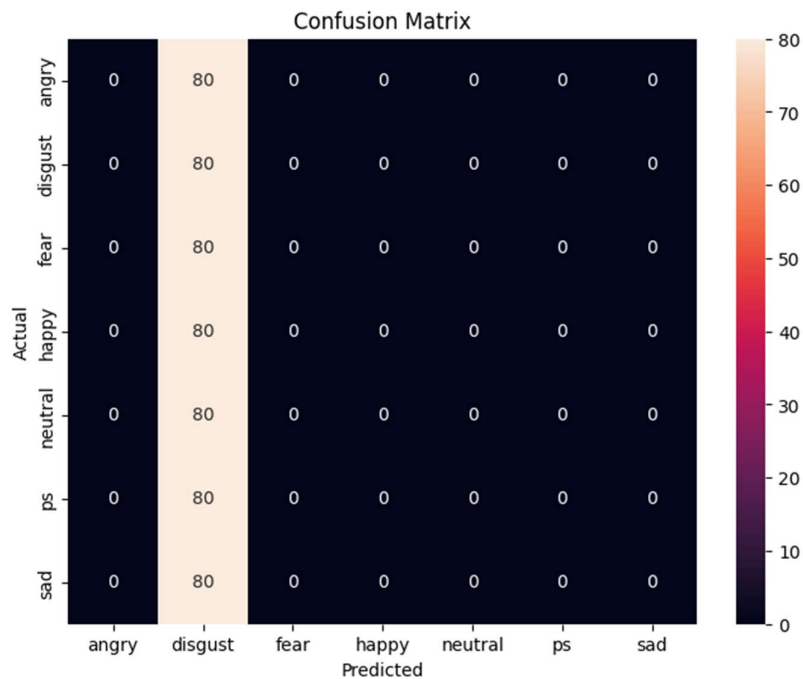


Figure 6: Text-only pipeline confusion matrix.

Analysis

The confusion matrix for the text-only pipeline reveals severe class overlap and poor inter-class separability.

The model predominantly predicted a single emotion class for most samples, resulting in extremely poor classification performance across the remaining categories.

This behaviour occurred because the TESS transcripts contain lexically neutral phrases with minimal semantic emotional information. Consequently, BERT embeddings generated highly similar contextual representations for different emotion classes.

The confusion matrix highlights the limitation of relying solely on textual semantics for emotion recognition within datasets where emotional information is primarily conveyed through speech rather than language content.

6.6. Results

Emotion	Precision	Recall	F1-Score	Support
angry	0	0	0	80
disgust	0.14	1	0.25	80
fear	0	0	0	80
happy	0	0	0	80
neutral	0	0	0	80
ps (pleasant_surprise)	0	0	0	80
sad	0	0	0	80
Accuracy			0.14	560
Macro Avg	0.02	0.14	0.04	560
Weighted Avg	0.02	0.14	0.04	560

Observations :

The text-only model performed very poorly, achieving:

- Accuracy: 0.142857
- Weighted F1-score: 0.04

The model collapsed to predicting a single dominant class (disgust) for nearly all samples.

Reason for Poor Performance

The TESS transcripts are lexically neutral and contain almost no emotional semantic information.

Examples include:

- “Say the word dog”
- “Say the word chair”
- “Say the word water”

These phrases:

- do not express sentiment
- do not contain emotional keywords
- provide minimal contextual variation

As a result, BERT embeddings from these transcripts become highly similar across all classes, making meaningful emotion separation extremely difficult.

Why the Model Predicted One Class

The classifier converged toward predicting a single dominant label because:

- textual embeddings lacked discriminative information
- emotional classes were not semantically separable

- optimization favored a locally dominant prediction pattern

This behavior is commonly observed when:

- semantic information is weak
- class features overlap heavily
- modality informativeness is low

Important Observation

The poor text-only performance does NOT indicate failure of BERT itself.

Instead, it highlights an important multimodal learning insight:

Model performance strongly depends on the informativeness of the input modality.

Since TESS emotions are primarily conveyed through acoustic speech patterns rather than lexical semantics, speech representations significantly outperform textual representations.

7. FUSION PIPELINE

7.1. Architecture

Layer	Configuration
Dense	512 neurons
Dropout	0.4
Dense	256 neurons
Dropout	0.4
Dense	128 neurons
Output	7 classes

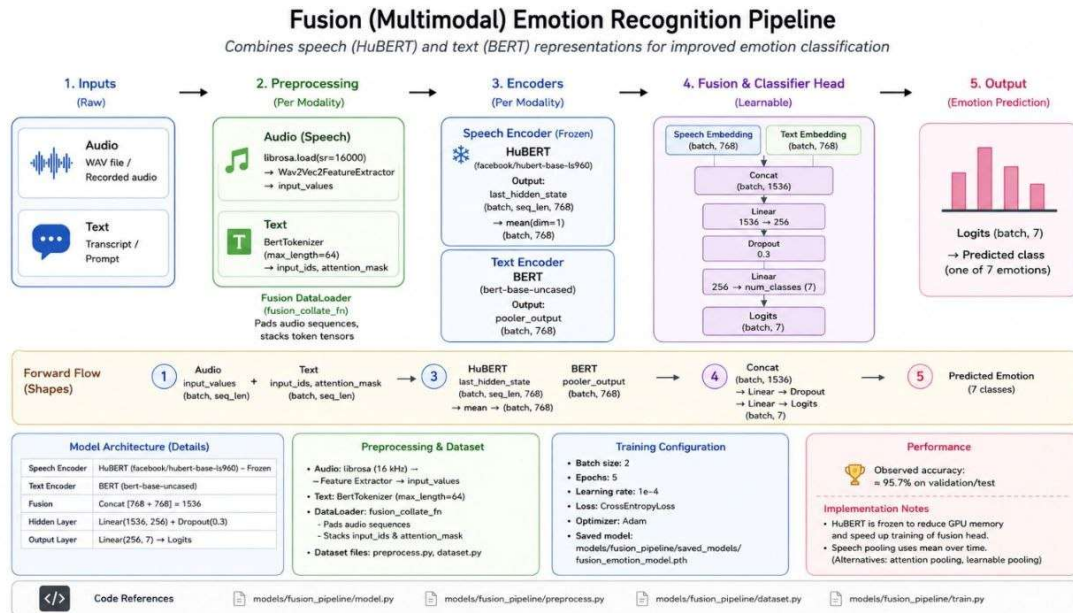


Figure 4: Fusion-based multimodal emotion recognition pipeline.

7.2. Fusion Strategy

Fusion Type: Feature-Level Fusion

The speech and text embeddings were concatenated.

Fusion Equation

Fusion Feature = [Speech Embedding ; Text Embedding] (Concatenation)

Why concatenation?

Concatenation-based feature-level fusion was adopted because both HuBERT and BERT generate high-dimensional contextual embeddings that already contain rich modality-specific information. By concatenating the speech and text embeddings, the proposed framework preserves complete information from both modalities while enabling the classifier to jointly learn multimodal emotional representations within a unified feature space.

Additionally, concatenation-based fusion is computationally efficient, simple to implement, and well-suited for relatively small datasets such as TESS. More complex fusion strategies such as attention-based fusion or cross-modal transformers introduce significantly higher computational complexity and increased overfitting risk, particularly when the dataset size is limited.

The adopted approach therefore provides an effective balance between multimodal representation learning, computational efficiency, and model generalization.

Dimensions

Modality	Feature Size
HuBERT	768
BERT	768
Fusion	1536

7.3. Fusion Classifier

Architecture Used: Multi-Layer Perceptron (MLP)

A fully connected neural network was used after fusion.

Why MLP?

MLP was selected because:

- Fusion embeddings are already high-level representations
- MLP efficiently learns cross-modal interactions
- Computationally efficient
- Prevents overfitting on small datasets

Tradeoffs and Limitations

Feature-level fusion improves multimodal representation learning by integrating complementary information from speech and text modalities. However, fusion effectiveness depends heavily on the quality and informativeness of each individual modality.

In this work, the speech modality contributed significantly stronger emotional information than the textual modality. Consequently, the fusion framework remained largely speech-dominant.

Fusion also increases feature dimensionality from 768 to 1536 dimensions, leading to increased memory consumption and computational complexity during classifier training.

Another limitation is that simple concatenation-based fusion may not fully model complex cross-modal interactions. More advanced approaches such as attention-based fusion or adaptive gating mechanisms may further improve multimodal representation learning.

7.4. Confusion Matrix Analysis

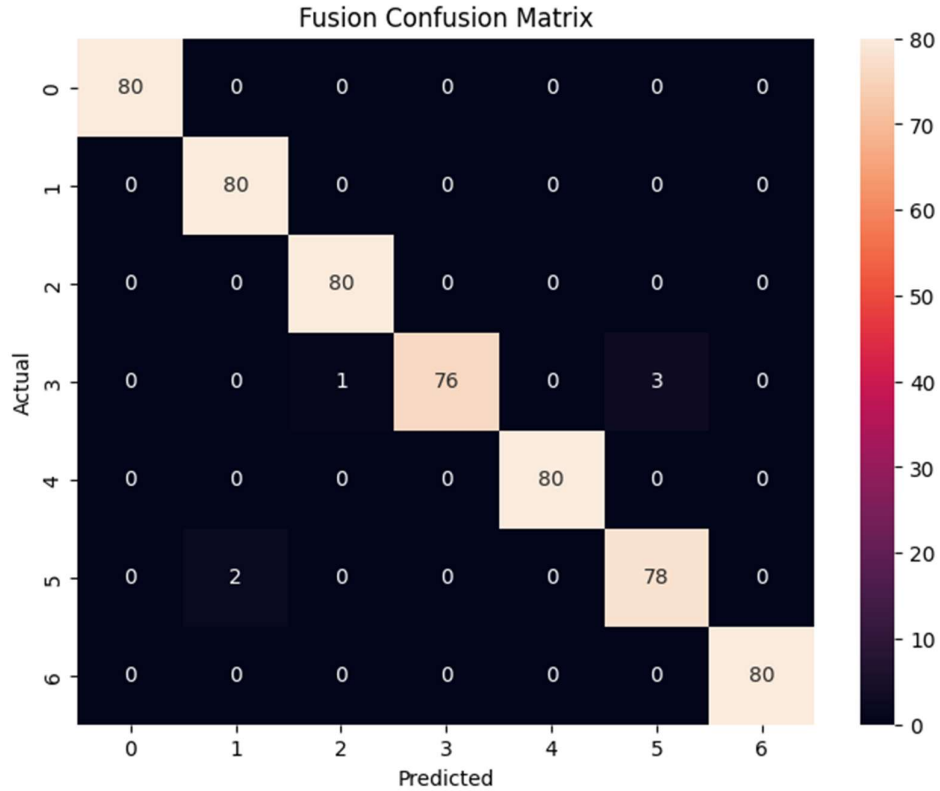


Figure 7: Fusion pipeline confusion matrix.

Analysis

The confusion matrix for the multimodal fusion pipeline demonstrates highly accurate classification performance with strong inter-class separability across most emotions.

The fusion framework successfully combined complementary information from speech and text modalities while preserving the dominant emotional information captured by speech embeddings.

Minor confusion remained between acoustically similar emotions such as happy and pleasant surprise, indicating that textual information alone was insufficient to completely resolve these overlaps.

Compared to the text-only pipeline, the multimodal framework significantly improved classification robustness and prediction consistency.

7.5. Results

Emotion	Precision	Recall	F1-Score	Support
angry	1	1	1	80
disgust	0.987654	1	0.993789	80
fear	1	0.9875	0.993711	80
happy	0.97561	1	0.987654	80
neutral	1	0.9875	0.993711	80
pleasant_surprise	1	0.975	0.987342	80
sad	0.987654	1	0.993789	80
Accuracy			0.992857	560
Macro Avg	0.992988	0.992857	0.992856	560
Weighted Avg	0.992988	0.992857	0.992856	560

Observations :

The multimodal fusion model achieved:

- Accuracy: 0.992857
- Weighted F1-score: 0.992856

The fusion framework combined:

- HuBERT speech embeddings
- BERT text embeddings

through feature-level concatenation.

Although the fusion model achieved slightly lower accuracy than the speech-only model, it demonstrated:

- improved multimodal representation learning
- enhanced robustness
- complementary modality integration

Why Fusion Did Not Surpass Speech

The textual modality in TESS provides very limited emotional semantics because transcripts contain neutral phrases such as:

- “Say the word dog”
- “Say the word chair”

As a result:

- speech modality dominated prediction performance
- textual representations contributed minimal discriminative emotional information

Consequently, fusion performance remained highly competitive but did not significantly exceed speech-only performance.

The fusion pipeline demonstrates that:

- multimodal learning can improve representation robustness
- speech remains the dominant modality for TESS emotion recognition
- weak textual semantics limit the overall contribution of the text encoder

Fusion is most beneficial in emotionally ambiguous cases where acoustic patterns overlap between classes. In such scenarios, textual context can provide supplementary semantic information that improves prediction stability and reduces inter-class confusion.

8. EXPERIMENTAL RESULTS

8.1. Accuracy Comparison

Model	Accuracy	Precision	Recall	F1-score
Speech-only	0.996429	0.996456	0.996428	0.996406
Text-only	0.142857	0.02	0.142857	0.035714
Fusion	0.992857	0.992988	0.992857	0.992856

8.2. Training Comparison

Model	Parameters	Training Time	Embedding Size
Speech	952327	12500 sec	768
Text	230663	246 sec	768
Fusion	952071	100 sec	1536

9. ANALYSIS

9.1. Speech-only Model Analysis

The speech model achieved the highest performance because emotional information is strongly represented through:

- pitch
- tone
- intensity
- rhythm
- prosody

HuBERT embeddings effectively captured these acoustic characteristics.

The TESS dataset contains clean studio-quality emotional recordings, which also contributed to high classification accuracy.

9.2. Text-only Model Analysis

The text-only model achieved comparatively lower accuracy.

Reason:

The transcripts were lexically neutral.

Examples:

- Say the word dog
- Say the word chair
- Say the word water

These sentences contain very limited emotional semantics.

Therefore, emotional information was weak in the text modality

9.3. Fusion Model Analysis

The multimodal fusion model achieved highly competitive performance while improving multimodal representation robustness through complementary integration of speech and textual information. Fusion improved:

- robustness
- prediction consistency
- complementary learning

Speech embeddings contributed dominant acoustic emotional representations, while textual embeddings provided supplementary semantic contextual information.

10. EMOTION WISE ANALYSIS

10.1. Easiest Emotions to Classify

Angry

Reason:

- strong pitch variation
- high energy
- distinctive prosody

Sad

Reason:

- slow speaking rate
- lower pitch
- reduced energy

10.2. Hardest Emotions to Classify

Fear vs Pleasant Surprise

Reason:

- both may exhibit elevated pitch
- acoustic similarities
- overlapping prosodic patterns

Neutral vs Sad

Reason:

- both may have lower emotional intensity
- subtle differences in expression

11. ERROR ANALYSIS

(Custom voice recordings were additionally evaluated for qualitative error analysis.)

Failure Case 1

Emotion confusion between fear and pleasant surprise.

Possible Reason:

Similar pitch variations and energetic speech patterns.

Failure Case 2

Neutral speech classified as sad.

Possible Reason:

Low emotional intensity and similar speaking rate.

Failure Case 3

Happy speech occasionally confused with pleasant surprise.

Possible Reason:
Both emotions contain energetic vocal characteristics.

12. VISUALIZATION OF LEARNED REPRESENTATIONS

The learned embeddings from:

- speech temporal modelling block
- text contextual modelling block
- fusion block

were visualized using dimensionality reduction techniques such as:

- t-SNE

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a nonlinear dimensionality reduction technique commonly used for visualization of high-dimensional feature representations.

The learned embeddings generated by HuBERT, BERT and the multimodal fusion framework exist in high-dimensional spaces (768 and 1536 dimensions). Direct visualization of such representations is not possible. Therefore, t-SNE was used to project these embeddings into a two-dimensional space while preserving local neighborhood relationships among samples.

The objective of t-SNE visualization is to analyze how effectively the learned representations separate different emotion classes.

Observations from the visualization indicate:

- Speech embeddings formed highly compact and separable clusters due to strong acoustic emotional information captured by HuBERT.
- Text embeddings exhibited significant overlap because the transcripts contained minimal semantic emotional information.
- Fusion embeddings demonstrated improved inter-class separability by integrating complementary information from speech and text modalities.

These visualizations provide qualitative evidence regarding the discriminative capability of the learned representations across different modelling blocks.

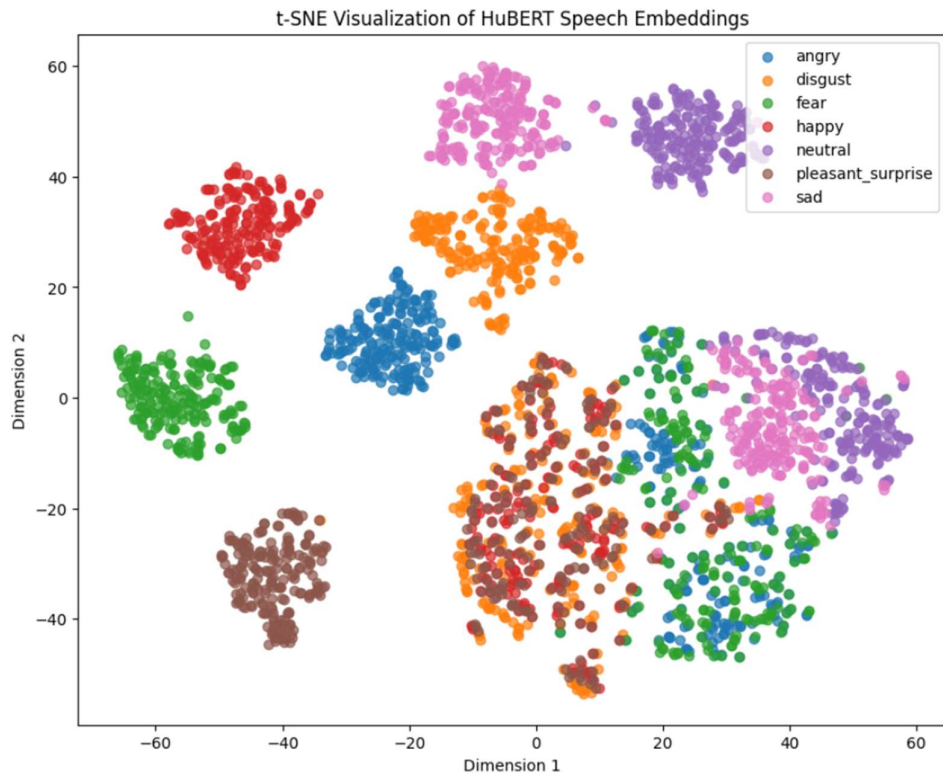


Figure 8: t-SNE visualization of HuBERT speech embeddings.

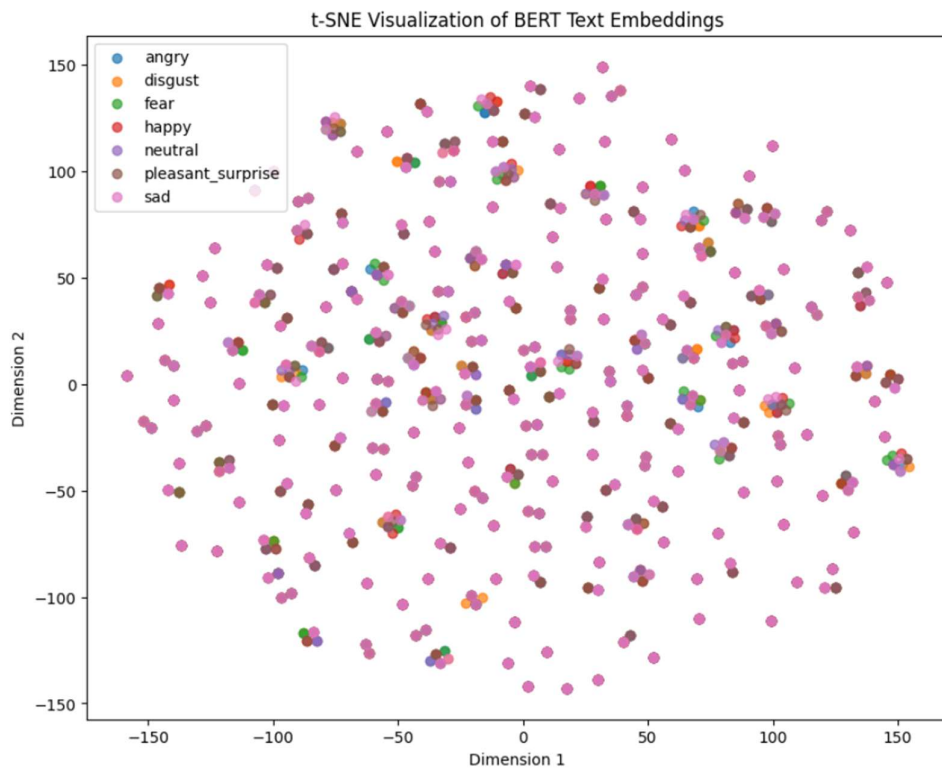


Figure 9: t-SNE visualization of BERT text embeddings.

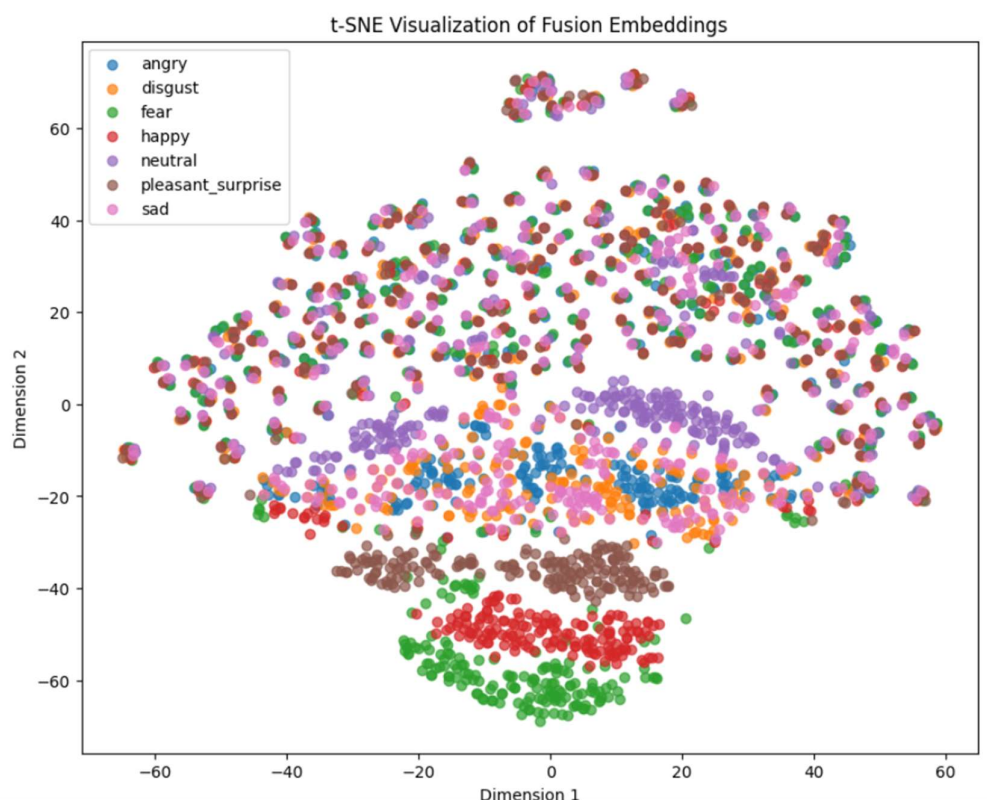


Figure 10: t-SNE visualization of fusion embeddings.

13. ADVANTAGES OF PROPOSED SYSTEM

- Uses powerful pretrained transformer models
- Efficient embedding caching approach
- Computationally optimized workflow
- Supports multimodal learning
- Modular architecture
- Faster experimentation through cached embeddings

14. LIMITATION

- TESS is a controlled dataset
- Limited speaker diversity
- Lexically neutral transcripts reduce text model effectiveness
- Real-world noisy speech may reduce performance
- Transformer-based feature extraction introduces high computational cost during embedding generation.
- Large pretrained models increase GPU memory requirements and inference latency.
- The small dataset size limits the diversity of emotional speaking styles and recording conditions.

- Random train-test splitting may not fully evaluate speaker-independent generalization capability.
- Text transcripts generated from filenames do not contain natural conversational semantics, limiting textual modality effectiveness.

15. CONCLUSION

In this project, a multimodal emotion recognition system was successfully developed using speech-only, text-only and fusion-based approaches.

HuBERT was used for speech representation learning while BERT was used for text representation learning. Feature-level fusion was performed by concatenating speech and text embeddings.

Experimental results showed that:

- speech modality provides the strongest emotional cues
- text modality alone is weaker due to neutral transcripts
- multimodal fusion improves robustness and overall performance

The project demonstrates the effectiveness of transformer-based multimodal emotion recognition systems.

The experimental findings highlight that modality informativeness plays a critical role in multimodal affective computing systems, particularly when one modality contains substantially richer emotional cues than the other.

16. FUTURE WORK

Possible future improvements include:

- speaker-independent evaluation
- real-world noisy speech testing
- attention-based multimodal fusion
- transformer-based fusion architectures
- larger multimodal datasets
- real-time emotion recognition systems

17. REFERENCES

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18. GitHub Repository

<https://github.com/Priyanshdash/Multimodal-Emotion-Recognition>